Lecture 24 (Weak and Strong Duality)

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Today we will formalize a few results related to duality.

Theorem (Weak Duality)

Suppose we have a minimization constraint in the primal. Suppose $\vec{b}$ is the right hand side vector, and $\vec{c}$ is the cost vector. Then, if $\vec{x}$ is a feasible solution to the primal LP and $\vec{p}$ is a feasible solution to the corresponding dual LP, then:

$$\vec{p} \cdot \vec{b} \leq \vec{c} \cdot \vec{x}$$

Proof

For any feasible $\vec{x}$, $\vec{p}$ for the primal and dual respectively, we can define

$$u_i = p_i \left(\vec{a}_i \cdot \vec{x} - b_i \right)$$

$$v_j = \left(c_j - \vec{p} \cdot A_j \right) x_j$$

where $\vec{a}_i$ is the ith row of A and A_j is the jth column of A. When converting to the dual, the sign of p_i will be the same as $(\vec{a}_i \cdot \vec{x} - b_i)$ (or if there is an equality constraint, then $(\vec{a}_i \cdot \vec{x} - b_i) = 0$). Similarly, x_j and $(c_j - \vec{p} \cdot A_j)$ have the same sign, thus

$$u_i, v_j \geq 0$$

Thus

$$0 \leq \sum_i u_i = \vec{p}^T \left(A\vec{x} - \vec{b} \right)$$

$$0 \leq \sum_j b_j = \left(\vec{c}^T - \vec{p}^T A \right) \vec{x}$$

Thus

$$0 \leq \sum_i u_i + \sum_j b_j = \vec{p}^T A\vec{x} - \vec{p}^T \vec{b} + \vec{c}^T \vec{x} - \vec{p}^T A\vec{x} = \vec{c}^T \vec{x} - \vec{p}^T \vec{b} = \vec{c} \cdot \vec{x} - \vec{p} \cdot \vec{b}$$

Thus

$$\vec{p} \cdot \vec{b} \leq \vec{c} \cdot \vec{x}$$

as desired.

Corollary 1

1. If the primal is unbounded (i.e. the optimal minimum is $-\infty$), then there does not exist a feasible solution to the dual.
2. If the dual is unbounded (i.e. the optimal maximum is ∞), then there does not exist a feasible solution to the primal.

Proof

Suppose the primal is unbounded and there exists a feasible solution, $\vec{p}$ to the dual. Then

$$C = \vec{p} \cdot \vec{b} \leq \vec{c} \cdot \vec{x}$$

for all feasible $\vec{x}$ in the LP. But as the primal is unbounded, then for any C we can find $\vec{x}$ such that

$$\vec{c} \cdot x < C$$

a contradiction! As such, the dual LP must be infeasible. A similar argument holds for part 2 of this corollary.

Corollary 2

If there exist feasible $\vec{x}^*$ in the primal LP and feasible $\vec{p}^*$ in the dual LP such that

$$\vec{p}^* \cdot \vec{b} = \vec{c} \cdot \vec{x}^*$$

then both $\vec{x}^*$ and $\vec{p}^*$ are optimal.

Proof

Suppose $\vec{x}$ and $\vec{p}$ are arbitrary feasible solution to the primal LP and dual LP respectively. Then by weak duality, we see

$$\vec{p} \cdot \vec{b} \leq \vec{c} \cdot \vec{x}^* = \vec{p}^* \cdot \vec{b} \leq \vec{c} \cdot \vec{x}$$

thus

$$\vec{p} \cdot \vec{b} \leq \vec{p}^* \cdot \vec{b}$$

and

$$\vec{c} \cdot \vec{x}^* \leq \vec{c} \cdot \vec{x}$$

for all feasible $\vec{p}, \vec{x}$.

Additional Notes

1. For MIPs, weak duality holds as well. The basic argument stems from the LP relaxation (this is where you replace each integer variable with a real variable with the same bounds to get an LP with a much larger feasible region). Since weak duality holds for all feasible $\vec{x}$, $\vec{p}$, then it must hold for the much smaller subset of integer valued vectors in the feasible region.
2. For LPs, weak duality implies a much stronger statement called, "Strong Duality". Strong duality doesn't apply to MIPs in general.

Theorem (Strong Duality)

If an LP has an optimal solution, then so does its dual. The optimal costs will be equal.

Proof

Consider the standard form

$$\text{minimize } \vec{c} \cdot \vec{x}$$

s.t.

$$A\vec{x} = \vec{b}$$

$$\vec{x} \geq 0$$

Suppose there exists an optimal solution to the above LP and the rows of A are linearly independent. Then we can apply simplex method to get an optimal vector, $\vec{x}$ with corresponding basis B. Let $\vec{x}_B = B^{-1}\vec{b}$. As $\vec{x}$ is optimal, then it must have non-negative reduced costs, thus

$$\vec{c}^T - \vec{c}^T B^{-1} A \geq 0$$

Suppose $\vec{p}$ is defined such that $\vec{p}^T = \vec{c}_B^T B^{-1}$. Then

$$\vec{c}^T - \vec{c}_B^T B^{-1} A = \vec{c}^T - \vec{p}^T A \geq 0$$

so

$$\vec{p}^T A \leq \vec{c}^T$$

And as the dual LP is

$$\text{maximize } \vec{p} \cdot \vec{b}$$

s.t.

$$\vec{p}^T A \leq \vec{c}^T$$

$$\vec{p} \in \mathbb{R}^m$$

then $\vec{p}$ will be a feasible solution to the dual LP. Now, consider the objectives:

$$\vec{p} \cdot \vec{b} = \vec{p}^T \vec{b} = \vec{c}_B^T B^{-1} \vec{b} = \vec{c}_B^T \vec{x}_B = \vec{c} \cdot \vec{x}$$

So by corollary 2, this implies both $\vec{x}$ and $\vec{p}$ are optimal.

Summary of Results

Strong duality implies that any equivalent LP or it's dual will all share the same optimal objective value is a finite optimum exists. We have a table below of all the possible cases below:

		The Primal		
		Has a Finite Optimum	Is Unbounded	Is Infeasible
The Dual	Has a Finite Optimum	Possible	–	–
	Is Unbounded	–	–	Possible
	Is Infeasible	–	Possible	Possible